19 Prove algebraically that the straight line with equation $x - 2y = 10$ is a tangent to the circle with equation $x^2 + y^2 = 20$

Rearrange the line: $y = \dfrac{x - 10}{2}$

Substitute: $x^2 + \dfrac{(x - 10)^2}{4} = 20$

1 mark
Substitute the line into the circle

Multiply by 4: $4x^2 + (x - 10)^2 = 80$

×	x	-10
x	x^2	$-10x$
-10	$-10x$	100

$4x^2 + x^2 - 10x - 10x + 100 = 80$

$5x^2 - 20x + 100 = 80$

1 mark
Expand the square correctly

$5x^2 - 20x + 100 = 80$
$\quad\quad -80 \quad\quad\quad -80$

$5x^2 - 20x + 20 = 0$
$\quad\quad \div 5 \quad\quad\quad \div 5$

1 mark
Three-term quadratic = 0

$x^2 - 4x + 4 = 0$

×	x	-2
x	x^2	$-2x$
-2	$-2x$	4

$(x - 2)^2 = 0$, so $x = 2$.

1 mark
Factorise to solve the quadratic

$y = (2 - 10) \div 2 = -4$

There is exactly one intersection, $(2, -4)$, so the line is a tangent.

1 mark
Complete algebra and the one-intersection conclusion

(Total for Question 19 is 5 marks)

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